INTRODUCTION TO APPLIED NUMERICAL METHODS

LU Decomposition

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LU DECOMPOSITION

Motivation

- It is inefficient to solve equations with the same coefficients [A] but with different right-hand-side constant: RHS (B) using Gauss Elimination
- Allows fast evaluation and solution for various [B] given a single value of [A]
- Is there a way to condense the Gauss elimination into a matrix form?

$$\begin{array}{ccccccccccc}
a_{11}x_1 & + & a_{12}x_2 & + & a_{13}x_3 & + & \cdots & + & a_{1n}x_n & = & b_1 \\
a_{21}x_1 & + & a_{22}x_2 & + & a_{23}x_3 & + & \cdots & + & a_{2n}x_n & = & b_2 \\
 & & \vdots & & \vdots & & & & & & \\
a_{n1}x_1 & + & a_{n2}x_2 & + & a_{n3}x_3 & + & \cdots & + & a_{nn}x_n & = & b_n
\end{array}$$

Operate on each single equation

$$\boxed{[A]\{X\} = \{B\}}$$

LU Decomposition

- Separate time-consuming elimination of the matrix [A] from the manipulation of the RHS [B]
- Once [A] is “manipulated”, multiple (different) RHS vectors can be evaluated in an efficient manner

LU DECOMPOSITION(CONT)

The system, $[A]\{x\}=\{b\}$, can be decomposed into two matrices

$[L]$: lower triangular matrix
$[U]$: upper triangular matrix

such that

$$[L][U]=[A]$$
$$[L][U]\{x\}=\{b\}$$

Set up the following systems

$$[L]\{d\}=\{b\}$$
$$[U]\{x\}=\{d\}$$

Step 1: $[L]\{d\}=\{b\}$ is used to generate an intermediate vector $\{d\}$ by forward substitution.
Step 2: $[U]\{x\}=\{d\}$ is used to get the answer, $\{x\}$, by back substitution.

LU DECOMPOSITION-BASED SOLUTION PROCESS

Process

- **Step 1**: LU Decomposition: [A] is factorized or “decomposed” into lower [L] and upper [U] triangular matrices, i.e., [A] = [L][U]
- **Step 2**: Substitution step: (i)

$$[L]\underbrace{[U]\{x\}}_{\{D\}}=\{B\}$$

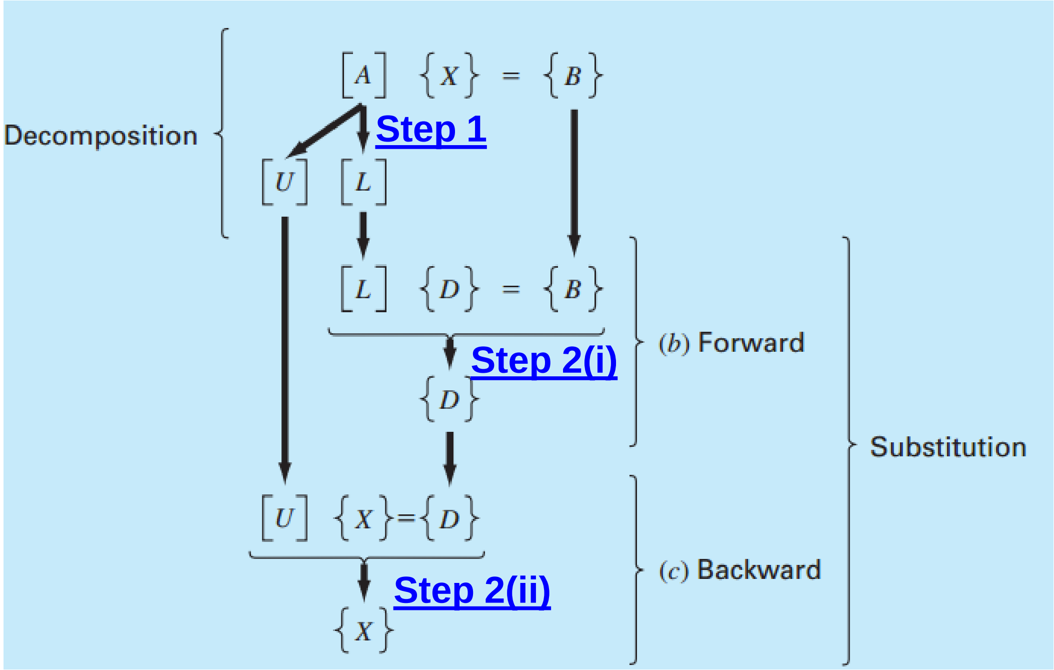
Introduce an intermediate vector {D}

$$\boxed{[L]\{D\} = \{B\}}$$

Forward Substitution

$$\boxed{\{D\}}$$

Backward Substitution (ii) $\boxed{[U]\{X\}=\{D\}}$



LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

[L] and [U] is produced from the row operation during forward elimination step

Step 1: Forward Elimination

Matrix/Vector Form

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix}$$

$$[A] : Row(2) - Row(1) \times \left(\frac{a_{21}}{a_{11}} \right)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Define: $f_{21} = \left(\frac{a_{21}}{a_{11}} \right)$

$$[A] : Row(3) - Row(1) \times \left(\frac{a_{31}}{a_{11}} \right)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

Define: $f_{31} = \left(\frac{a_{31}}{a_{11}} \right)$

LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

[L] and [U] is produced from the row operation during forward elimination step

$\downarrow$

Matrix/Vector Form

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

$$\downarrow \quad [A]: Row(3') - Row(2') \times \left(\frac{a'_{32}}{a'_{22}} \right)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

Define: $f_{32} = \left(\frac{a'_{32}}{a'_{22}} \right)$

LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

Key idea: use f_{21}, f_{31}, f_{32}, … to assemble the lower triangular matrix [L]

$$\boxed{[L] = \begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix}}$$

$$\boxed{[U] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}} \quad \Rightarrow \quad [A] \to [L][U]$$

$$\begin{array}{lll} l_{11} = 1 & l_{12} = 0 & l_{13} = 0 \\ l_{21} = f_{21} & l_{22} = 1 & l_{23} = 0 \\ l_{31} = f_{31} & l_{32} = f_{32} & l_{33} = 1 \end{array}$$

$$\begin{array}{lll} u_{11} = a_{11} & u_{12} = a_{12} & u_{13} = a_{13} \\ u_{21} = 0 & u_{22} = a'_{22} & u_{23} = a'_{23} \\ u_{31} = 0 & u_{32} = 0 & u_{33} = a''_{33} \end{array}$$

LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

Process

- **Step 2(i):** Forward substitution step to obtain {D}

$$[L]\{D\} = \{B\}$$

$$d_i = b_i - \sum_{j=1}^{i-1} l_{ij} d_j \quad \text{for } i = 2,3,\ldots,n$$

$$\begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

- **Step 2(ii):** Back-substitution step to obtain {x}

$$[U]\{X\} - \{D\} = 0$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix}$$

$$x_n = \frac{d_n}{u_{nn}}$$

$$x_i = \frac{d_i - \sum_{j=i+1}^{n} u_{ij} d_j}{u_{ii}} \quad \text{for } i = n-1, n-2, \ldots, 1$$

AN EXAMPLE

Derive an LU decomposition based on the Gauss elimination using the previous example

$$
\begin{aligned}
3x_1 - 0.1x_2 - 0.2x_3 &= 7.85 \\
0.1x_1 + 7x_2 - 0.3x_3 &= -19.3 \\
0.3x_1 - 0.2x_2 + 10x_3 &= 71.4
\end{aligned}
$$

Find the solution to x based on LU decomposition

Solution: **The matrix form of the equation set**

$$
[A] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.1 & 7 & -0.3 \\ 0.3 & -0.2 & 10 \end{bmatrix}
$$

Following forward elimination, the upper triangular matrix [U] was

$$
[U] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix}
$$

Determine the lower triangular matrix [L]: need factors for forward elimination (see next several slides)

$$
f_{21} = \frac{0.1}{3} = 0.03333333 \qquad f_{31} = \frac{0.3}{3} = 0.1000000 \qquad f_{32} = \frac{-0.19}{7.00333} = -0.0271300
$$

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AN EXAMPLE: FORWARD ELIMINATION

Use Gauss elimination to solve

$$[A] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ \boxed{0.1} & 7 & -0.3 \\ 0.3 & -0.2 & 10 \end{bmatrix} \quad \begin{matrix} (1) \\ (2) \\ (3) \end{matrix}$$

Solution: **Forward Elimination**

Step 1.1: Row (1)X(0.1/3)

$$① \times \left(\frac{0.1}{3}\right)$$

$$\boxed{f_{21} = \left(\frac{0.1}{3}\right)}$$

$$\begin{bmatrix} 0.1 & -\frac{0.01}{3} & -\frac{0.02}{3} \end{bmatrix} \quad (1^*)$$

Row (2) – Row (1*)

$$\begin{bmatrix} 0.1 & 7 & -0.3 \end{bmatrix} \quad (2)$$

$$-$$

$$\begin{bmatrix} 0.1 & -\frac{0.01}{3} & -\frac{0.02}{3} \end{bmatrix} \quad (1^*)$$

$$\Downarrow$$

$$\begin{bmatrix} 0 & 7.00333 & -0.293333 \end{bmatrix} \quad (2') \Rightarrow$$

New A Matrix

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0.3 & -0.2 & 10 \end{bmatrix}$$

AN EXAMPLE: FORWARD ELIMINATION

A matrix

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ \boxed{0.3} & -0.2 & 10 \end{bmatrix} \begin{array}{l} (1) \\ (2') \\ (3) \end{array}$$

$$f_{31} = \left(\frac{0.3}{3}\right)$$

Solution: **Forward Elimination**

Step 1.2: Row (1)X(0.3/3)

$$1 \times \left(\frac{0.3}{3}\right)$$

$$\boxed{\begin{bmatrix} 0.3 & -0.01 & -0.02 \end{bmatrix}} \quad (1^+)$$

Row (3) – Row (1+)

$$\boxed{\begin{bmatrix} 0.3 & -0.2 & 10 \end{bmatrix}} \quad (3)$$

$$-$$

$$\boxed{\begin{bmatrix} 0.3 & -0.01 & -0.02 \end{bmatrix}} \quad (1^+)$$

$$\Downarrow$$

$$\boxed{\begin{bmatrix} 0 & -0.19 & 10.02 \end{bmatrix}} \quad (3') \Rightarrow$$

New A Matrix

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & -0.19 & 10.02 \end{bmatrix} \begin{array}{l} (1) \\ (2') \\ (3') \end{array}$$

AN EXAMPLE: FORWARD ELIMINATION

A Matrix

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & \boxed{-0.19} & 10.02 \end{bmatrix} \quad \begin{matrix} (1) \\ (2') \\ (3') \end{matrix}$$

Solution: **Forward Elimination**

Step 2:

Row (2')X(-0.19/7.00333)

$$\boxed{\begin{bmatrix} 0 & -0.19 & 0.0079581 \end{bmatrix}}$$

$$2' \times \left(-\frac{0.19}{7.00333}\right)$$

(2*)

$$f_{32} = \left(-\frac{0.19}{7.00333}\right)$$

Row (3') - Row (2*)

$$\boxed{\begin{bmatrix} 0 & -0.19 & 10.02 \end{bmatrix}} \quad (3')$$

$$-$$

$$\boxed{\begin{bmatrix} 0 & -0.19 & 0.0079581 \end{bmatrix}} \quad (2^*)$$

$$\Downarrow$$

$$\boxed{\begin{bmatrix} 0 & 0 & 10.012 \end{bmatrix}} \quad (3'') \Rightarrow$$

New A Matrix

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.012 \end{bmatrix}$$

AN EXAMPLE

$$\begin{aligned} 3x_1 - 0.1x_2 - 0.2x_3 &= 7.85 \\ 0.1x_1 + 7x_2 - 0.3x_3 &= -19.3 \\ 0.3x_1 - 0.2x_2 + 10x_3 &= 71.4 \end{aligned}$$

Solution:

Step 3: Determine the lower triangular matrix [L]: need factors for forward elimination

$$f_{21} = \frac{0.1}{3} = 0.03333333 \qquad f_{31} = \frac{0.3}{3} = 0.1000000 \qquad f_{32} = \frac{-0.19}{7.00333} = -0.0271300$$

$$\boxed{[L] = \begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix}}$$

The LU decomposition is:

$$\boxed{[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix}}$$

AN EXAMPLE

Solution:

Step 4: [A] = [L][U] ????

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix}$$

verify

$$[L][U] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.0999999 & 7 & -0.3 \\ 0.3 & -0.2 & 9.99996 \end{bmatrix}$$

AN EXAMPLE

Solution:

Step 5: A forward-substitution to compute {D}: $\boxed{[L]\{D\} = \{B\}}$

$$d_i = b_i - \sum_{j=1}^{i-1} L_{ij} d_j \quad \text{for } i = 2, 3, \ldots, n$$

Forward substitution

$$\begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} 7.85 \\ -19.3 \\ 71.4 \end{Bmatrix}$$

$$\left.\begin{aligned} d_1 &= 7.85 \\ d_2 &= -19.3 - 0.0333333(7.85) = -19.5617 \\ d_3 &= 71.4 - 0.1(7.85) + 0.02713(-19.5617) = 70.0843 \end{aligned}\right\} \quad \boxed{\{D\} = \begin{Bmatrix} 7.85 \\ -19.5617 \\ 70.0843 \end{Bmatrix}}$$

AN EXAMPLE

Solution:

Step 6: A backward-substitution to compute {X}: $\boxed{[U]\{X\}=\{D\}}$

$$x_n = \frac{d_n}{u_{nn}}, \quad x_i = \frac{d_i - \sum_{j=i+1}^{n} u_{ij} d_j}{u_{ii}} \quad \text{for } i = n-1, n-2, \ldots, 1$$

Back-substitution

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 7.85 \\ -19.5617 \\ 70.0843 \end{Bmatrix}$$

$$\{X\} = \begin{Bmatrix} 3 \\ -2.5 \\ 7.00003 \end{Bmatrix}$$